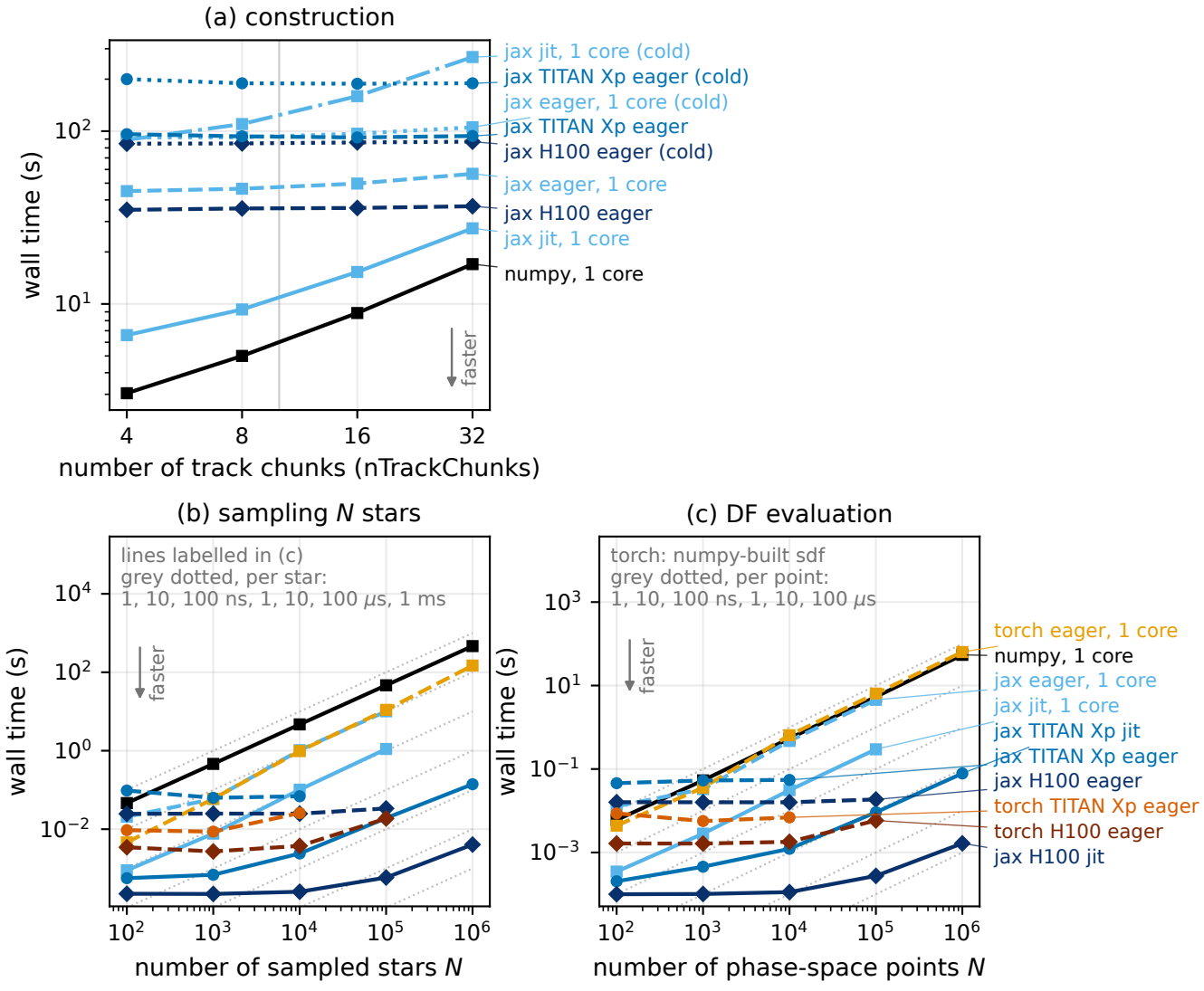
(a) construction
wall time (s)
10^2
10^1
faster
4
8
16
32
number of track chunks (nTrackChunks)
jax jit, 1 core (cold)
jax TITAN Xp eager (cold)
jax eager, 1 core (cold)
jax TITAN Xp eager
jax H100 eager (cold)
jax eager, 1 core
jax H100 eager
jax jit, 1 core
numpy, 1 core
(b) sampling N stars
lines labelled in (c)
grey dotted, per star:
1, 10, 100 ns, 1, 10, 100 μs, 1 ms
wall time (s)
10^4
10^2
10^0
10^{-2}
faster
10^2
10^3
10^4
10^5
10^6
number of sampled stars N
(c) DF evaluation
torch: numpy-built sdf
grey dotted, per point:
1, 10, 100 ns, 1, 10, 100 μs
wall time (s)
10^3
10^1
10^{-1}
10^{-3}
faster
10^2
10^3
10^4
10^5
10^6
number of phase-space points N
torch eager, 1 core
numpy, 1 core
jax eager, 1 core
jax jit, 1 core
jax TITAN Xp jit
jax TITAN Xp eager
jax H100 eager
torch TITAN Xp eager
torch H100 eager
jax H100 jit